Revision energy — Study guide

Grade 6 · IB MYP 1 Sciences · Unit 1 Thinking about Energy · whole Physics unit

Name: _____ Class: _____ Date: _____

How to use this study guide

- It helps you revise the whole Energy unit before your **test**.
- Write your answers in the **My answer** column. Use full sentences and science words.
- When you check your work with the answer sheet, write what you got right (✔️) or need to fix in the **Self-check** column.
- Stuck on a word? Use the **Energy words glossary** (10 languages): scan the QR code or go to school.taneski.com/science/g06

Energy glossary

What I need to know — checklist

Tick each goal you are confident about now. Come back and tick the rest after you finish the study guide.

- ☐ I can define energy and state its unit, the joule (J).
- ☐ I can name 9 types of energy and give an example of each.
- ☐ I can state the law of conservation of energy.
- ☐ I can describe energy transfers and say which energy is useful and which is wasted.
- ☐ I can draw and read a Sankey diagram.
- ☐ I can explain charge and current, outline how electricity is made, and draw circuit symbols and circuits.
- ☐ I can explain the difference between heat and temperature, and describe solids, liquids and gases using particles.
- ☐ I can explain conduction, convection and radiation, and give examples of conductors and insulators.
- ☐ I can compare renewable and non-renewable energy resources and suggest ways to save energy.

A · Energy and its stores

#	What I need to know (question or situation)	My answer	Self-check
A1	**Define** energy. **Explain** why energy is hard to think about.		
A2	**State** the unit of energy. Read the label of one of your favourite foods: how much energy does it have in 100 g?		
A3	**List** 9 types (stores) of energy and give an example of each.		
A4	**State** the law of conservation of energy. **Explain** what it means for energy transfers.		
A5	*Situation:* A skateboarder stands at the top of a ramp and then rolls down. **Describe** the energy transfers.		
A6	*Situation:* You wind up a toy and let it go. **Describe** where the energy comes from and where it goes.		

Type of energy	Example

B · Energy transfers and efficiency

#	What I need to know (question or situation)	My answer	Self-check
B1	**Give** 3 examples of energy transfers. Name the types of energy in each one.		
B2	**Describe** the useful and the wasted energy for a light bulb and a hair dryer.		
B3	**Describe** how to draw a Sankey diagram. Then **draw** one on the grid for a light bulb: 100 J of electrical energy in, 20 J of light out, 80 J of heat out. What fraction of the energy is useful?		
B4	*Situation:* A phone charger gets warm while it charges a phone. **Explain** what happens to the energy.		

Device	Energy in	Useful energy out	Wasted energy out
light bulb			
hair dryer			

C · Electricity

#	What I need to know (question or situation)	My answer	Self-check
C1	**List** three things you use every day that need electricity.		

#	What I need to know (question or situation)	My answer	Self-check
C2	**Define** electrical energy and charge.		
C3	**Define** electric current.		
C4	**Outline** how electricity is made in a power station.		
C5	**Draw** the circuit symbol of each component and **state** what it does.		

Component	Symbol	What it does
cell		
battery		
lamp (bulb)		
switch		
wire		
buzzer		
motor		

C6 **Describe** how to draw a circuit diagram. Then **draw** a circuit with a cell, a switch and a lamp.

my circuit

#	What I need to know (question or situation)	My answer	Self-check
C7	*Situation:* You build a circuit, but the lamp does not light. **Suggest** three reasons.		

D · Heat and matter

#	What I need to know (question or situation)	My answer	Self-check
D1	**Describe** how heat and temperature are different. Include the unit of each.		
D2	**State** what particles make up matter.		
D3	**Describe** the three states of matter. Draw the particles in each box.		
D4	**Explain** what happens to the particles when matter is heated and when it is cooled.		
D5	**Describe** which way heat moves between two objects at different temperatures.		

	Solid	Liquid	Gas
Particles (draw)			
How they are arranged			
How they move			
Shape			
Volume			

<table>
<tr><th>#</th><th>What I need to know (question or situation)</th><th>My answer</th><th>Self-check</th></tr>
<tr><td>D6</td><td colspan="2">Explain how heat is transferred by conduction, convection and radiation.<table><tr><th></th><th>How it works</th><th>Where it happens</th><th>Example</th></tr><tr><td>Conduction</td><td></td><td></td><td></td></tr><tr><td>Convection</td><td></td><td></td><td></td></tr><tr><td>Radiation</td><td></td><td></td><td></td></tr></table></td><td></td></tr>
<tr><td>D7</td><td>State the difference between conductors and insulators. Give two examples of each.</td><td></td><td></td></tr>
<tr><td>D8</td><td>Situation: A cooking pan is made of metal but its handle is made of plastic. Explain why.</td><td></td><td></td></tr>
<tr><td>D9</td><td>Situation: A radiator is on one side of a cold room. Explain how the whole room gets warm.</td><td></td><td></td></tr>
</table>

E · Energy resources

#	What I need to know (question or situation)	My answer	Self-check
E1	**Describe** four ways to save energy at home.		

<table>
<tr><th>#</th><th>What I need to know (question or situation)</th><th>My answer</th><th>Self-check</th></tr>
<tr><td>E2</td><td colspan="2">List energy resources and give an example of where each is used. Tick if it is renewable.<table><tr><th>Energy resource</th><th>Example of use</th><th>Renewable? ✔️</th></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td></tr></table></td><td></td></tr>
<tr><td>E3</td><td>List three fossil fuels.</td><td></td><td></td></tr>
<tr><td>E4</td><td>Describe the difference between renewable and non-renewable energy resources.</td><td></td><td></td></tr>
<tr><td>E5</td><td colspan="2">State one advantage and one disadvantage of each energy resource.<table><tr><th>Resource</th><th>Advantage</th><th>Disadvantage</th></tr><tr><td>fossil fuels</td><td></td><td></td></tr><tr><td>nuclear</td><td></td><td></td></tr><tr><td>wind</td><td></td><td></td></tr><tr><td>solar</td><td></td><td></td></tr><tr><td>hydroelectric</td><td></td><td></td></tr></table></td><td></td></tr>
<tr><td>E6</td><td>Situation: A small sunny and windy town needs a new power station. Suggest which energy resource it should choose. Explain your choice with one advantage and one disadvantage.</td><td></td><td></td></tr>
</table>

F · Quick check

Circle the best answer: A, B, C or D.

F1 What type of energy is stored in food?

- (A) kinetic
- (B) chemical
- (C) sound
- (D) nuclear

F2 In a coal power station, which order of energy changes is right?

- (A) chemical → thermal → kinetic → electrical
- (B) electrical → kinetic → thermal → chemical
- (C) thermal → chemical → electrical → kinetic
- (D) kinetic → chemical → thermal → electrical

F3 The particles in a solid are …

- (A) far apart and moving fast
- (B) close together and sliding past each other
- (C) close together in rows and vibrating in place
- (D) far apart and not moving

F4 A metal bench feels colder than a wooden bench at the same temperature. Why?

- (A) the metal is colder
- (B) metal is a good conductor: it takes heat away from your hand quickly
- (C) wood makes its own heat
- (D) metal gives out cold

F5 Why do double-glazed windows reduce heat loss?

- (A) the glass is coloured
- (B) the air between the two panes is an insulator
- (C) glass is a good conductor
- (D) the windows are always closed

F6 Energy cannot be …

- (A) stored
- (B) transferred
- (C) created or destroyed
- (D) measured

F7 Water is heated in a kettle. How does the heat spread through the water?

- (A) conduction
- (B) convection
- (C) radiation
- (D) insulation

F8 Which material is the best conductor of heat?

- (A) plastic
- (B) wood
- (C) air
- (D) copper

F9 In which heat transfer do particles pass on energy by vibrating, without moving from place to place?

- (A) conduction
- (B) convection
- (C) radiation
- (D) evaporation

F10 Which of these makes electricity from moving air?

- (A) a solar panel
- (B) a wind turbine
- (C) a coal power station
- (D) a battery

Self-reflection

Question	My answer
Two things I am confident about	
Two things I still need to revise	
One situation question I found hard, and why	
My plan to revise before the test	